

HEBA SLEIMAN

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Education

M2 MoSIG — Master of Science in Informatics, Artificial Intelligence specialization

2026 – 2027 (In Progress)

Université Grenoble Alpes / Grenoble INP, Grenoble, France

B.S. Computer Science, Minor in Data Analytics — Graduated with Honors

May 2025

Lebanese American University, Beirut, Lebanon

Experience

Research Intern — Real-Time Ergonomic Risk Assessment in VR (RULA), G-SCOP Lab, Université Grenoble Alpes

Jan – Jun 2026

Grenoble, France — Supervised by Dr. Mathieu Delangle

- Built an end-to-end closed-loop system combining motion capture, real-time classification, and automatic workstation adaptation (see Projects for details)

Software Engineering Intern — simBrAln Solutions

Jul 2024

Beirut, Lebanon

- Performed data preprocessing and feature engineering, then developed and integrated ML models into production web applications
- Built full-stack features — email routing, real-time notifications, interactive UI — using ASP.NET MVC and Bootstrap

Projects

Real-Time Ergonomic Risk Assessment in VR (RULA) — Master's Research Internship, G-SCOP Lab, Université Grenoble Alpes, supervised by Dr. Mathieu Delangle (Python, scikit-learn, Unreal Engine 5, OptiTrack, TCP sockets)

- Built an end-to-end closed-loop system that captures motion data using OptiTrack and Meta Quest, computes joint angles in Unreal Engine, generates a labeled dataset, serves a trained classifier over a Python TCP server, and delivers real-time risk visualization with automatic workstation adaptation
- Trained and compared Random Forest and SVM classifiers on 6 joint angles per side, selecting Random Forest for its stronger recall on high-risk postures, and built a proof-of-concept loop where the workstation's table height adapts automatically once a tracked joint enters a high-risk angle range

Resume-to-JD Matching Agent (Python, Anthropic Claude API, Pydantic, Streamlit)

- Built an LLM pipeline using Claude tool-use / function calling to extract structured candidate profiles from resume PDFs and score them against a job description
- Implemented a bias-audit module that re-scores candidates with demographic signals swapped and flags score drift, plus an eval harness for probabilistic LLM output
- Deployed the agent publicly on Render with a Streamlit interface

Tasklet Support Agent (Python, Anthropic Claude API, SQLite)

- Built a multi-turn conversational support agent using Claude tool-use, with per-user data scoping enforced at the service layer rather than by the LLM
- Ran adversarial refusal red-teaming (jailbreak attempts, fabricated authority, injected fake instructions) to validate the agent's refusal behavior held under pressure

Financial Sentiment Classification with Transformers — Course project, M2 MoSIG (Python, PyTorch, HuggingFace Transformers)

- Fine-tuned DistilBERT on the Financial PhraseBank dataset (3.4k sentences, 3 imbalanced classes), raising performance from 0.74 accuracy / 0.60 macro-F1 (TF-IDF + Naive Bayes baseline) to **0.94 accuracy / 0.93 macro-F1**
- Diagnosed via confusion matrices why a frozen-encoder model failed (1/84 negative sentences detected, confused with positive) and fixed it by full fine-tuning with a low learning rate, reaching 93% recall on the negative class

Publications

Numerical Solutions of the Nonlinear Poliovirus Model: A Radial Basis Dual Layered Neural Network Approach

Biomedical Signal Processing and Control, 2025 — with Dr. Zulqurnain Sabir, M.A. Abdelkawy & Fatima Hassouna

- Co-developed a dual-layer radial basis function neural network (18 and 34 neurons, trained via Bayesian regularization) to numerically solve a nonlinear four-compartment (susceptible-infected-exposed-vaccinated) poliovirus transmission model
- Validated solver accuracy against a Runge-Kutta-generated dataset using error histograms, regression coefficients, and training-performance diagnostics

Technical Skills

- Programming:** Python, Java, C, SQL, HTML, CSS, PHP
- Machine Learning:** scikit-learn, Random Forest, SVM, Logistic Regression, Gradient Boosting, fairness metrics (PPV, FPR, FNR), calibration analysis, AUC-ROC
- Deep Learning & NLP:** PyTorch, HuggingFace Transformers, BERT/DistilBERT fine-tuning, tokenization, TF-IDF
- Generative AI & LLMs:** Anthropic Claude API, LLM tool use / function calling, prompt engineering, structured-output validation (Pydantic), LLM evals & red-teaming
- Simulation & Real-Time Systems:** Unreal Engine 5, OptiTrack motion capture, TCP sockets
- Data & Visualization:** pandas, Tableau, SQLite, Oracle SQL, data modeling
- Tools & Deployment:** Git/GitHub, Streamlit, Render, ASP.NET MVC
- Languages:** Arabic (native), English (fluent)

References

Dr. Mathieu Delangle — G-SCOP Lab, Université Grenoble Alpes (RULA project supervisor)

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